

Solution for Q.no. 117 and 118:

117. Option (b) is correct.

$$\Sigma x_i f_i = 1.1 + \frac{2}{2} + \frac{3}{2^2} + \dots + \frac{n}{2^{(n-1)}} \dots (i)$$

$$\text{and } \frac{1}{2} \Sigma x_i f_i = \frac{1}{2} + \frac{2}{2^2} + \dots + \frac{n-1}{2^{n-1}} + \frac{n}{2^n} \dots (ii)$$

Eq (i) - (ii) we get

$$\Rightarrow \left(1 - \frac{1}{2}\right) \Sigma x_i f_i = \left[1.1 + \frac{2-1}{2} + \frac{3-2}{2^2} + \frac{4-3}{2^3} + \dots + \frac{1}{2^{n-1}}\right] - \frac{n}{2^n}$$

$$\Rightarrow \frac{1}{2} \Sigma x_i f_i = \left[1 + \frac{1}{2} + \frac{1}{2^3} + \frac{1}{2^4} + \dots + \frac{1}{2^{n-1}}\right] - \frac{n}{2^n}$$

$$\Rightarrow \frac{1}{2} \Sigma x_i f_i = \frac{1 \left(1 - \frac{1}{2^n}\right)}{1 - \frac{1}{2}} - \frac{n}{2^n}$$

$$= \frac{2(2^n - 1)}{2^n} - \frac{n}{2^n}$$

$$\Rightarrow \Sigma x_i f_i = \frac{2^{n+1} - n - 2}{2^{n-1}}$$

118. Option (c) is correct.

$$\Sigma f_i = 1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{(n-1)}}$$

$$\Sigma f_i = \frac{\left(1 - \frac{1}{2^n}\right)}{1 - \frac{1}{2}} = \frac{2^n - 1}{2^{n-1}}$$

$$\text{Now, Mean} = \frac{\Sigma f_i x_i}{\Sigma f_i}$$

$$= \frac{2^{n+1} - n - 2}{2^{n-1}} \cdot \frac{2^{n-1}}{2^n - 1}$$

$$= \frac{2^{n+1} - n - 2}{2^n - 1}$$

Solution for Q.no. 119 and 120:

Ascending data is, 15, 18, 19, 21, 24, 32, 35, 41, 47, 50

Largest 5 observations are, 32, 35, 41, 47, 50

$$\text{Mean } \mu = \frac{32 + 35 + 41 + 47 + 50}{5}$$

$$= \frac{205}{5}$$

$$\Rightarrow \mu = 41$$

119. Option (c) is correct.

$$\text{Mean deviation} = \frac{\Sigma |x_i - \mu|}{n}$$

$$= \frac{|32 - 41| + |35 - 41| + |41 - 41| + |47 - 41| + |50 - 41|}{5}$$

$$= \frac{9 + 6 + 0 + 6 + 9}{5}$$

$$= 6$$

120. Option (d) is correct.

$$\sigma^2 = \frac{\Sigma (x_i - \mu)^2}{5}$$

$$= \frac{9^2 + 6^2 + 0^2 + 6^2 + 9^2}{5}$$

$$= \frac{234}{5}$$

$$= 46.8$$